

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)
Innovative Practices Description

UNIT I ANALOG MODULATION

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Amplitude Modulation

Name of the Innovative Practice: Lecture Video

Date & Duration: 09-07-2019 & 15 minutes

ICT Tool Used:- Desktop, Projector, Smart Class room

Description of Lecture Video:

A video lecture is a video which presents educational material for a topic which is to be learned. It might be a video of a teacher speaking to the camera, photographs and text about the topic or some mixture of these.

Goals(Learning Outcomes)

- The Students will be able to transmit the signal using AM technique.
- The Students will be able to design AM communication Systems

Use of appropriate methods:

Justification for choosing the topic using Lecture Video:

Studies have shown that the use of short video clips allows for more efficient processing and memory recall. The visual and auditory nature of videos appeals to a wide audience and allows each user to process information in a faster way to them.

The basic communication starts from here. One must know about how to modulate and demodulate the signal. So I chose this topic as a lecture video for better understanding and how the signal transmits from sender to receiver.

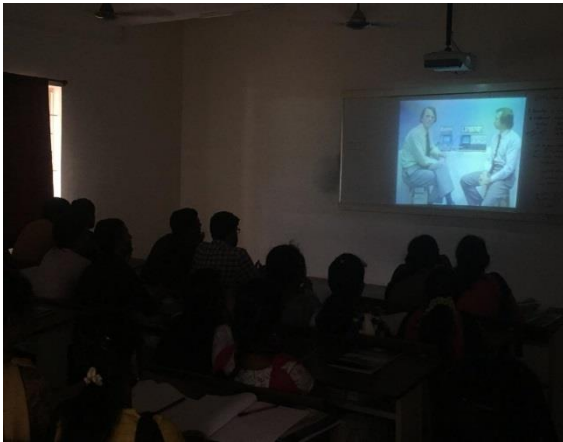
Effective Presentation

Details of Implementation

The Students watched the entire lecture video of how to modulate and demodulate the signal using AM technique.

Execution

MIT Lecture Video on Amplitude Modulation



MIT Professors gave Demonstration Lecture on Amplitude Modulation



Students watching MIT Lecture Video

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The success of the activity was evaluated by asking the AM modulation and demodulation in Internal Assessment test I – Around 90% of students answered correctly.

Reflective Critique:

Benefits of conducting this activity

After Completion of the video, students gave feedback that they got a clear idea about how to transmit the low frequency signal over a long distance, and they asked me to conduct more such activities like this.

Challenges

For better Clarity, I repeat some portion of the videos again to watch. Also the video quality was not much good.

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	1			1	2	2	2	2

CO1: The Students will be able to design AM and Angle modulated Communication systems

References:

1. H Taub, D L Schilling, G Saha, “Principles of Communication Systems” 3/e, TMH 2007
2. <https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/video-lectures/lecture-14-demonstration-of-amplitude-modulation/>

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UNIT II WAVEFORM CODING

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Sampling of signals

Name of the Innovative Practice: Laboratory Demonstration

Date & Duration: 25-07-2019 & 15 minutes

ICT Tool Used:- Communication Systems Laboratory

Description of Laboratory Demonstration:

The purpose of lab demonstration is to help the students learn how the signal was modulated and transmitted. Lab demonstration involves teaching and explanation. If a student is having problems with their execution you should try to discover where their understanding is in error and provide an explanation that will allow the student to achieve a correct understanding. The range of problems experienced by students is large and varied and this is one of the reasons that lab demonstration is interesting.

Goals(Learning Outcomes)

- The Students will be able to know how the continuous time signal is converted into discretized signal
- The Students will be able to describe about the concepts of sampling

Use of appropriate methods:

Justification for choosing Simulation:

Students studied theoretically how the sampled signal should be. He/She studied the concepts of how to convert the continuous message signal into sampled signal. But in the lab class only he/she can able to view the signal physically. Also he/she got a clear idea about how the continuous time signal is mixed with discrete carrier pulse signal to produce the sampled signal.

Effective Presentation

Details of Implementation

Students demonstrated the Sampling of signals using Communication System Lab trainer kit

Execution

Demonstration of Sampling of signals using Communication Lab Trainer Kit



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The assessment of effectiveness of the activity was felt when asked questions after demonstration

Reflective Critique:

Benefits of conducting this activity

- The laboratory experiences will be more likely to achieve the goals with clear learning outcomes in mind.
- The laboratory demonstration incorporates ongoing student reflection and discussion.
- The laboratory demonstrations are thoughtfully sequenced into the flow of classroom instruction.
- As they studied the concepts as a physical appearance, they were well versed in the particular topic.

Challenges

NIL

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	1	1				1		2		2

CO2: The students will be able to describe various pulse modulation techniques

References:

1. <https://www.youtube.com/watch?v=yWqrx08UeUs>
2. <https://nptel.ac.in/content/storage2/courses/108101039/download/Lecture-28.pdf>

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UNIT III DIGITAL MODULATION AND TRANSMISSION

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Phase Shift Keying – Digital Modulation

Name of the Innovative Practice: Simulation using MATLAB

Date & Duration: 16-08-2019 & 20 minutes

ICT Tool Used:- Desktop, MATLAB tool, Communication Systems Laboratory

Description of Simulation:

Simulation is the imitation of the operation of a real-world process or system over time. The act of simulating something first requires that a model be developed; this model represents the key characteristics or behaviors of the selected physical or abstract system or process. The model represents the system itself, whereas the simulation represents the operation of the system over time.

Goals(Learning Outcomes)

- The Students will be able to know how the signal was modulated and transmitted.
- The Students will be able to change the parameter values and see what happens

Use of appropriate methods:

Justification for choosing Simulation:

Simulation allows students to change parameter values and see what happens. I chose this particular topic as a MATLAB simulation, because students studied the same concepts in theory. So they got better idea while implementing the same using MATLAB. Due to this activity, the students were able to simulate the original information into modulated data using delta modulation

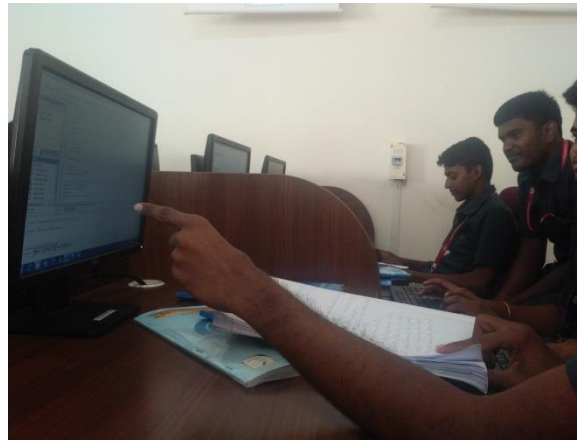
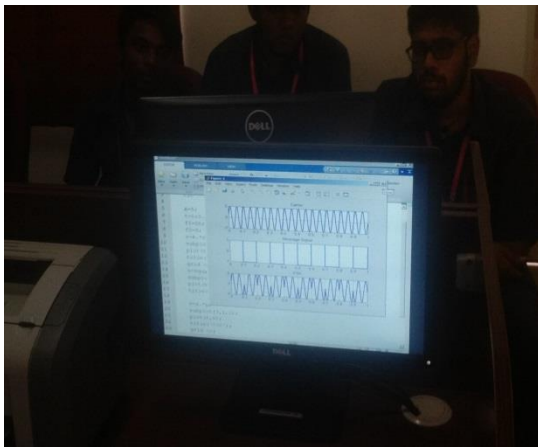
Effective Presentation

Details of Implementation

Students simulated the PSK modulation and demodulation using MATLAB tool.

Execution

Students Simulated PSK signal using MATLAB Simulation Tool



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The assessment of effectiveness of the activity was felt when asked this question in Internal Assessment test. Around 80% of the students answered correctly.

Reflective Critique:

Benefits of conducting this activity

After Completion of Simulation, the laboratory experiences will be more likely to achieve the goals with clear learning outcomes in mind. As they studied this concepts as a physical appearance, they were well versed in the particular topic. The laboratory Simulations are thoughtfully sequenced into the flow of classroom instruction.

Challenges

Some students felt little bit difficult to understand the MATLAB codes. But after my explanation about MATLAB commands, they got clear idea.

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	2	2	1	1			1		2	1	1

CO3: The students will be able to apply digital modulation techniques in various applications like Satellite, mobile & DTH Services.

References:

1. <https://in.mathworks.com/help/comm/ref/pskmod.html>
2. <https://in.mathworks.com/matlabcentral/fileexchange/44822-matlab-code-for-psk-modulation-and-demodulation?focused=3805515&tab=function>
3. S. Haykin, "Digital Communications", John Wiley, 2005

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UNIT IV INFORMATION THEORY AND CODING

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Convolution Coding

Name of the Innovative Practice: Animated Video

Date & Duration: 20-09-2019 & 15 minutes

ICT Tool Used:- Smart Class Room

Description of Animated Video:

Animated video is an engaging instrument that can help you tell your story more comprehensively. 65% of people are visual learners, and 90% of the information transmitted to the brain is visual. Animation evokes emotions that resonate with the viewers. It sparks conversations and makes your audience want to follow the story.

Goals(Learning Outcomes)

- The Students will be able to encode and decode a continuous stream of bits
- The Students will be able to design the system to detect, prevent and correct the errors during transmission.

Use of appropriate methods:

Justification for choosing this topic as animated video:

Studies have shown that the use of short video clips allows for more efficient processing and memory recall. The visual and auditory nature of videos appeals to a wide audience and allows each user to process information in a faster way to them. So I chose this topic as a lecture video, because it is important to know about how to encode and decode the continuous stream of bits. Because of the nature of convolution coding technique the binary bit stream source is convolved by applying some binary operations on them.

Effective Presentation

Details of Implementation

The Students watched the animated video of Convolution coding about how to encode and decode the signal.

Execution

Students watched Animated video on “Convolutional Coding”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The success of the activity was evaluated by asking this topic in Internal Assessment test III Examinations. Most of them were answered correctly.

Reflective Critique:

Benefits of conducting this activity

After Completion of the video, students gave feedback that they got a clear idea about how to encode and decode the message using Convolution coding, and they asked me to conduct more such activities like this.

Challenges

For better Clarity, I replayed the video again as by the student requisition.

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	3	2	2					1	1		1

CO4: The Students will be able to analyse Source and Error control coding

References:

1. S. Haykin, "Digital Communications", John Wiley, 2005
2. J.G Proakis, "Digital Communication", 4th Edition, Tata Mc Graw Hill Company, 2001
3. <https://www.youtube.com/watch?v=kRIfpmiMCpU>
4. <https://www.youtube.com/watch?v=dKIf6mQUfnY&t=1008s>

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UNIT V SPREAD SPECTRUM AND MULTIPLE ACCESS

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: EC8395 Communication Engineering

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Multiple Access Techniques

Name of the Innovative Practice: Role Play

Date & Duration: 17-10-2019 & 20 minutes

ICT Tool Used:- Smart Class Room

Description of Animated Video:

Role-play is any speaking activity when you either put yourself into somebody else's shoes, or when you stay in your own shoes but put yourself into an imaginary situation. It is widely agreed that learning takes place when activities are engaging and memorable.

Goals(Learning Outcomes)

- The Students will be able to analyze the different types of multiple access techniques.
- The Students will be able to know about how the mobile communication happens.

Use of appropriate methods:

Justification for choosing this topic as Role Play:

Incorporating role-play into the classroom adds variety, a change of pace and opportunities for a lot of language production and also a lot of fun! It can be an integral part of the class and not a 'one-off' event. If the teacher believes that the activity will work and the necessary support is provided, it can be very successful. Also one must know about how multiple users access the channel simultaneously and communicate between sender and receiver.

Effective Presentation

Details of Implementation

The Students performed the role play activity on the topic of Multiple Access Techniques.

Execution

Students performed Role play on “Multiple Access Techniques”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The success of the activity was evaluated by asking this topic in Internal Assessment test III Examinations. Most of them were answered correctly.

Reflective Critique:

Benefits of conducting this activity

After Completion of the activity, students gave feedback that they got a clear idea about the difference between multiple access techniques, and they asked me to conduct more such activities like this.

Challenges

NIL

Changes required for future (if required):

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	1	2					1	2	1	1

CO5: The Students will be able to explain Spread Spectrum and Multiple Access techniques

References:

1. <https://study.com/academy/lesson/role-play-method-of-teaching-definition-benefits.html>
2. <https://www.assessmentcentrehq.com/assessment-centre-exercises/role-plays-the-ultimate-guide/>
3. H Taub, D L Schilling, G Saha, “Principles of Communication Systems” 3/e, TMH 2007
4. S. Haykin “Digital Communications” John Wiley 2005